IT Workshop II (Cloud Computing): Hadoop - Concepts

October 7, 2026

Apache Pig - Combining Data

- grunt> DUMP A;
  (2,3)
  (1,2)
  (2,4)
- grunt> DUMP B;
  (z,x,8)
  (w,y,1)
- C = UNION A,B;
  grunt> DUMP C;
  (2,3)
  (1,2)
  (2,4)
  (z,x,8)
  (w,y,1)

User Defined Functions

- Pig provides extensive support for user defined functions (UDFs) as a way to specify custom processing. Pig UDFs can currently be implemented in six languages: Java, Jython, Python, JavaScript, Ruby and Groovy.
- The most extensive support is provided for Java functions. You can customize all parts of the processing including data load/store, column transformation, and aggregation.
- **Writing Java UDFs:**
  – myscript.pig
  REGISTER myudfs.jar;
  A = LOAD 'student_data' AS (name: chararray, age: int, gpa: float);
  B = FOREACH A GENERATE myudfs.UPPER(name);
  DUMP B;
- The command below can be used to run the script.
  pig -x local myscript.pig

User Defined Functions

- Let us now look at the implementation of the UPPER UDF.

```
package myudfs;
import java.io.IOException;
import org.apache.pig.EvalFunc;
import org.apache.pig.data.Tuple;

public class UPPER extends EvalFunc<String>
{
  public String exec(Tuple input) throws IOException {
      if (input == null || input.size() == 0 || input.get(0) == null)
           return null;
       try{
           String str = (String)input.get(0);
          return str.toUpperCase();
       }catch(Exception e){
           throw new IOException("Caught exception processing input row ", e);
       }
   }
 }
```

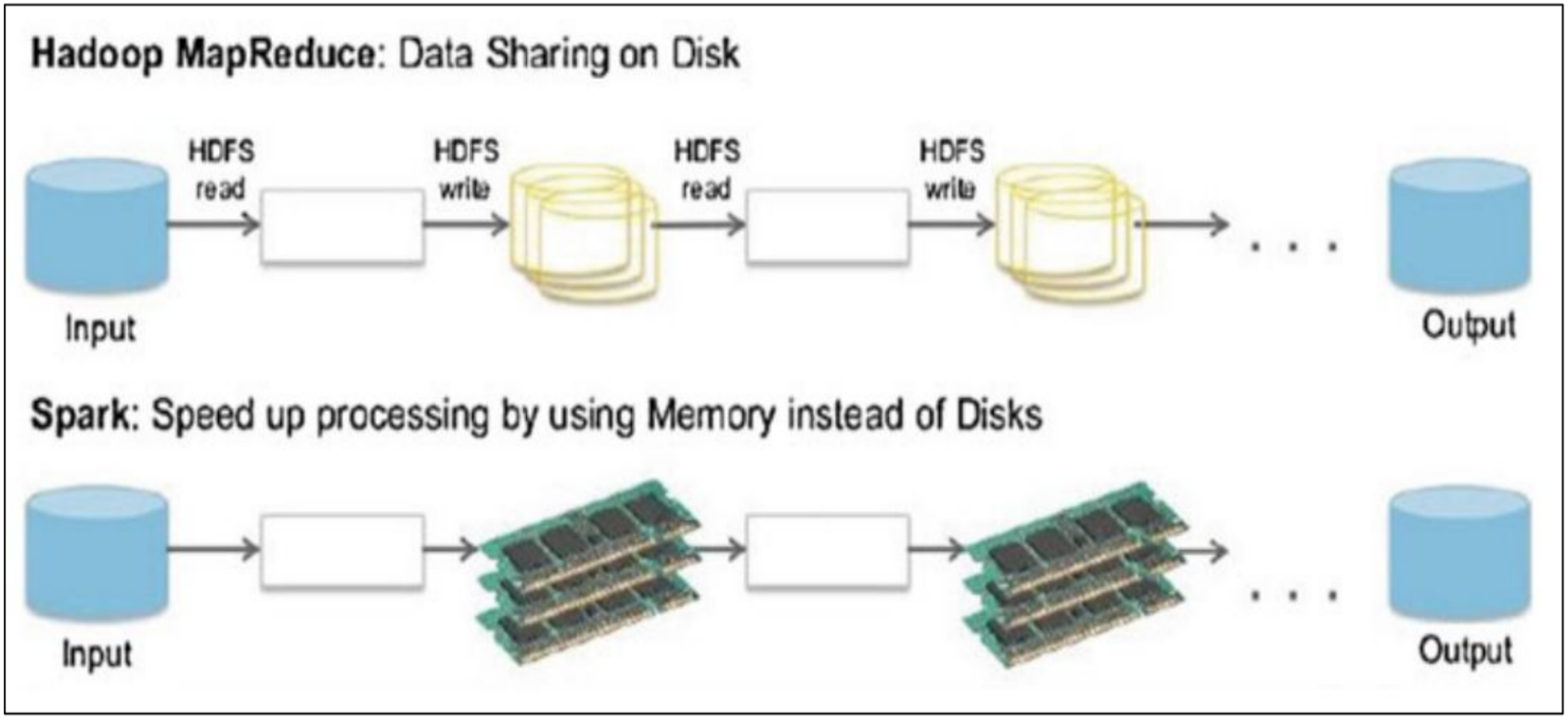
Apache Spark - Introduction

- Apache Spark is a cluster computing framework for large-scale data processing.
- Spark does not use MapReduce as an execution engine; instead it uses its own distributed runtime for executing work on a cluster.
- However, Spark has many parallels with MapReduce, in terms of both API and runtime.
- Spark is closely integrated with Hadoop: it can run on YARN and works with Hadoop file formats and storage backends like HDFS.

Apache Spark - Introduction

- Spark is best known for its ability to keep large working datasets in memory between jobs.
- This capability allows Spark to outperform the MapReduce workflow, where datasets are always loaded from disk.
- Two types of applications that benefit greatly from Spark's processing model are iterative algorithms and interactive analysis.
- Spark's user experience is better as it has rich set of APIs for performing many common data processing tasks, such as joins.
- Spark has native Scala, Java, Python, and R support.

Spark Uses Memory Instead of Disk



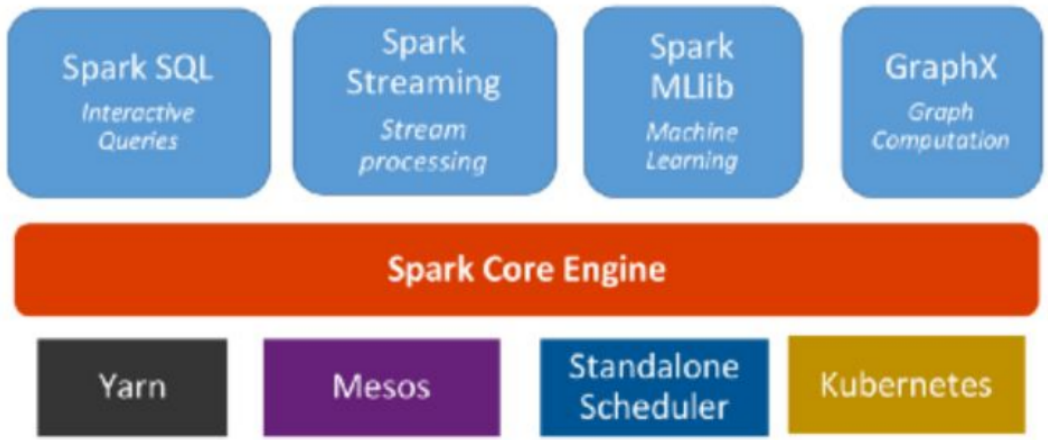
Apache Spark Architecture

Spark

Unified, open source, parallel, data processing framework for Big Data Analytics

Spark Unifies:

- Batch Processing
- Real-time processing
- Stream Analytics
- Machine Learning
- Interactive SQL



The End